



MicroRNAs in the developmental toolbox — a comparative approach to understanding their role in regulating insect development

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MicroRNAs are ubiquitous in the genomes of metazoans. Since their discovery during the late 20th century, our understanding of these small, noncoding RNAs has grown rapidly. However, there are still many unknowns about the functional significance of miRNAs — especially in non-model insects. Here I discuss the accumulating evidence that microRNAs are part of gene regulatory networks that determine not only the developmental outcome but also mediate transitions between stages and alternative developmental pathways. During the last 20 years, researchers have published a multitude of profiling studies that describe changes in miRNAs that may be important for development and catalog potential targets. Proof-of-principle studies document phenotypic changes that occur when candidate genes and/or miRNAs are inhibited or overexpressed. Studies that use both of these approaches, along with methods for confirming miRNA–mRNA interaction, demonstrate the necessary roles for miRNAs within gene networks. Together, all of these types of studies provide essential clues for understanding the function of miRNAs in the developmental toolbox.

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Introduction

MicroRNAs (miRNAs) are one class of small noncoding RNAs that regulate protein synthesis by limiting the translation of target genes. They were first described by researchers investigating the mechanisms that regulate

developmental timing in *Caenorhabditis elegans*. Soon after miRNAs were discovered, researchers realized that miRNAs are ubiquitous gadgets, if you will, among the many tools (*i.e.*, molecular complexes and regulatory pathways) that mediate cell cycle progression, cell growth and differentiation, developmental timing, and other processes involved in turning a genotype into a phenotype. miRNAs are evolutionarily widespread and are part of developmental programs in animals ranging from sponges to mammals [1–3]. Not only is the presence of miRNAs ubiquitous in metazoan genomes, but many miRNA sequences are highly conserved [3]. As a class of regulators, the function of miRNAs also seems to be evolutionarily conserved in numerous biological processes. However, whether the function of any particular miRNA sequence is also conserved remains a conspicuous knowledge gap — especially for non-model insects [4,5].

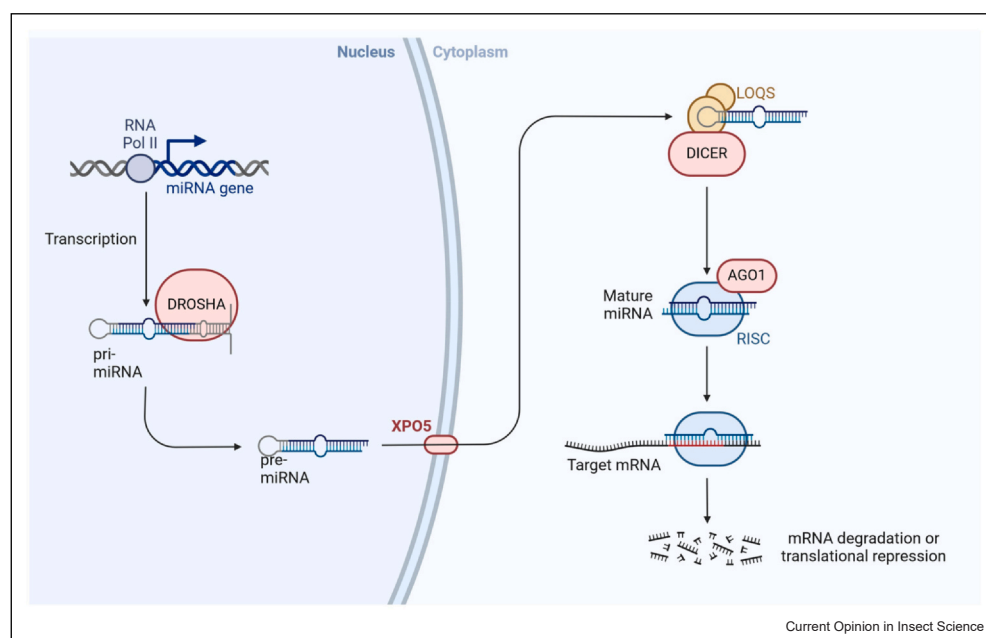
During the last decade, the number of published studies on miRNAs in non-model insects has increased exponentially. Improvements in experimental methods and approaches have made it easier to identify miRNAs and investigate their function in insects [5,6]. As a result, studies on planthoppers, aphids, bees, silk moths, locusts, and other insects are beginning to reveal new insights into miRNA function in general, especially during development. My goal for this review is to provide an overview of recent studies that demonstrate the function of miRNAs in specific pathways or processes. I also discuss what we are beginning to understand about the role of miRNAs as part of gene regulatory networks (GRNs). Finally, I suggest some necessary next steps for further understanding the function of miRNAs in the developmental toolbox.

What do we currently understand about microRNAs and insect development?

MicroRNAs are necessary for achieving the correct developmental outcome

Mature miRNAs are produced through a series of steps beginning with the transcription of a long, primary miRNA (pri-miRNA) and ending with a short (18–23 nucleotide) duplex of mature miRNA strands. One strand of the duplex binds to the RNA-induced silencing complex (RISC), guides the RISC to the target sequence, and interferes with the translation of the transcript (Figure 1) [1,2]. Silencing one or more of the genes that encode the miRNA machinery (e.g. Dicer1,

Figure 1



miRNA biogenesis begins in the nucleus with transcription of a primary miRNA (pri-miRNA) by RNA Polymerase II. The enzyme Drosha cleaves the pri-miRNA to produce a miRNA precursor (pre-miRNA) that is exported to the cytoplasm. Additional processing by Dicer creates a duplex with two mature microRNAs. One of the mature miRNAs binds to Argonaute, the catalytic center of the RISC, and guides it to a target mRNA. Binding of the RISC to the target blocks its translation and/or initiates degradation. Figure created using BioRender.com.

Argonaute, Drosha, or Pasha) is associated with developmental defects including, but not limited to, developmental arrest, failure to molt, malformed appendages, and deformed ovaries or male accessory glands. Defects that occur after silencing one or more components of miRNA biosynthesis have been described for whitefly (*Bemisia tabaci*), cotton bollworm (*Helicoverpa armigera*), silkworm (*Bombyx mori*), brown planthopper (*Nilaparvata lugens*), fruit fly (*Drosophila melanogaster*), and cockroach (*Blattella germanica*), and others [7–11]. Studies with *B. mori* [10,12] and *B. germanica* [11,13] provide more evidence of causality.

Mutants of *B. mori* that lack Dicer1 (i.e., *BmDicer1*-Knockout [KO] mutants) stop feeding during the 3rd larval instar, fail to molt, and eventually die. The observed defects have been linked to widespread differences in the availability of mature miRNAs in *BmDicer1*-KO mutants compared to their wild-type counterparts [10]. For example, let-7, one miRNA that is downregulated in *BmDicer1*-KO individuals, has been experimentally linked to developmental defects. Knockdown of let-7 in larvae of *B. mori* causes the same developmental effects caused by knocking out Dicer1 [12]. Although the pathways that are interrupted by removing Dicer1 and/or let-7 have not been identified, these studies demonstrate a link between Dicer1, let-7, and successful completion of the larval stage of development.

In *B. germanica*, using RNA interference (RNAi) to silence Dicer1 during the last nymphal instar prevents molting to the adult stage. Silencing Dicer1 reduces the availability of miRNAs in general and miR-2 in particular. miR-2 targets Krüppel homolog 1 (Kr-h1), a transcription factor that needs to be depleted before metamorphosis can occur. Without sufficient levels of miR-2, Kr-h1 remains at levels that are too high to initiate metamorphosis, and nymphs are not able to molt to the adult stage. Injecting synthetic miR-2 into nymphs lacking Dicer1 is sufficient to reduce Kr-h1 to a level that allows metamorphosis to proceed [11,13].

MicroRNA expression is dynamic, and fluctuations are necessary for correct outcomes

Distinct, stage-specific miRNA profiles have been described for cockroaches (*Periplaneta americana*), beetles (e.g., *Leptinotarsa decemlineata*, *Diabrotica virgifera*, and *Tribolium castaneum*), butterflies (*Bicyclus anynana*), mosquitoes (*Anopheles stephensi*), and flies (*Bactrocera dorsalis*) [5,14–20]. Dynamic changes in miRNA abundance can also occur within a developmental stage, as described in miRNA profiling studies with female adults of *Culex pipiens* mosquitoes and pupae of *Pieris napi* [21,22]. Many of these studies describe the dynamic nature of miRNA expression during development, list putative gene targets, and make inferences about the significance of the observed changes in

miRNA profiles. Although these studies do not demonstrate a definitive role for miRNAs, they are valuable for identifying biologically relevant changes in abundance and developing follow-up studies. A study with the red flour beetle, *Tribolium castaneum*, demonstrates a link between dynamic miRNA expression and development. In *T. castaneum*, changes in miR-8-3p are important for the normal development of eyes, wings, legs, and ovaries or male accessory glands. miR-8-3p is highly expressed during late larval stages, downregulated during the pupal stage, and then upregulated soon after adults emerge. Depleting miR-8-3p causes developmental defects in eyes, wings, legs, and ovaries or male accessory glands. *Eyegone*, a confirmed target of miR-8-3p, is highly expressed during the pupal stage. Increasing miR-8-3p abundance with a miR-8 mimic leads to an inappropriate decrease in *Eyegone* protein as well as defects in eye size or pigmentation [23]. Other studies that demonstrate the significance of dynamic miRNA expression focus on hormone signaling in the context of molting and metamorphosis, which are discussed below.

MicroRNAs are necessary for transitions between developmental stages

In addition to their role in morphology and other developmental outcomes, miRNAs regulate the timing of transitions between developmental stages (e.g. larva-to-pupa and pupa-to-adult transitions in holometabolous insects and nymph-to-adult transitions in hemimetabolous insects). Studies with *B. mori*, *Helicoverpa armigera*, and *Locusta migratoria* demonstrate the necessity of specific miRNA-mRNA interactions for successful molting or metamorphosis [24–27].

The insect-specific miRNA miR-14 regulates the larval-to-pupal transition in *Bombyx mori* and *Helicoverpa armigera* [24–26]. In *B. mori*, miR-14 regulates ecdysone synthesis and activity. Manipulating miR-14 abundance slows development and delays molting [24,25]. Two confirmed targets of miR-14-3p, Ecdysone receptor (EcR) and Early response gene E75, regulate the ecdysone signal but do not change ecdysone availability. Changing the abundance of miR-14-3p with inhibitors or mimics delays molting [24]. Increasing the abundance of miR-14-5p with a miRNA mimic also slows development [25]. In addition to targeting EcR and Early response gene E74, miR-14-5p regulates two proteins in the ecdysone biosynthesis pathway, Neverland and Shadow. Inhibitors and mimics of miR-14-5p disrupt development by affecting ecdysone titers and the downstream response [25].

In *H. armigera*, miR-14, along with miR-2766 and miR-277, regulates larva-to-larva and larva-to-pupa transitions by controlling the output of dopamine, a neurotransmitter known to be important for molting in *H. armigera* and other insects [26,27]. miR-14 and miR-2766 negatively regulate tyrosine hydroxylase (TH), a rate-limiting enzyme in the dopamine biosynthesis pathway. Increasing miR-14 or

miR-2766 with miRNA mimics decreases both TH protein and transcript abundance and prevents pupation. miR-277 also limits dopamine production by preventing the translation of Dopa Decarboxylase, an enzyme downstream of TH in the dopamine synthesis pathway. Treating larvae with miR-277 mimics leads to a decrease in Dopa Decarboxylase availability and prevents pupation [27]. Together these studies demonstrate that miRNAs are important regulators of dopamine biosynthesis in *H. armigera*. Experimentally inhibiting dopamine synthesis also causes defects in molting and metamorphosis in *D. melanogaster* and *Antheraea pernyi* [28,29]. However, it is not currently known whether miR-14, miR-2766, or miR-277 regulate dopamine titers in any species other than *H. armigera*.

MicroRNAs regulate metamorphosis in a microRNA/Kruppel/juvenile hormone feedback loop

In *Locusta migratoria*, metamorphosis is regulated through a feedback loop that includes Krüppel homolog 1 (Kr-h1), juvenile hormone (JH), let-7, and miR-278 [30,31]. Kr-h1 is a JH-induced transcription factor that must be depleted before metamorphosis can occur. let-7 and miR-278 target Kr-h1 and reduce the output of the protein sufficiently to initiate metamorphosis. Treating nymphs of *L. migratoria* with mimics of let-7 and miR-278 during the penultimate instar leads to precocious metamorphosis. Kr-h1 is also necessary for production of yolk protein, ovary development, and oocyte maturation in adults of *L. migratoria*. Injecting mimics of let-7 and miR-278 into adult females reduces vitellogenin production and causes ovaries to be smaller than those in the control group [31].

Similar interactions between miRNAs and Kr-h1 are found in *Blatella germanica* and *D. melanogaster* [11,32]. In *B. germanica*, *kr-h1* is regulated by miR-2 rather than let-7 and miR-278. However, the action of miR-2 (i.e. limiting Kr-h1 and promoting the transition from nymph to adult) is conserved [11]. In *D. melanogaster*, miR-927 regulates *kr-h1* [32]. A strain of *D. melanogaster* that overexpresses miR-927 in the fat body has slower rates of larval development, delayed pupation, and reduced fecundity compared to wild-type flies. Developmental defects and delays caused by miR-927 overexpression are rescued in flies that also overexpress *kr-h1*, as well as in flies treated with the JH analog Methoprene. Methoprene reduces miR-927 and increases Kr-h1 titers, which demonstrates a link between JH, miR-927, and Kr-h1. Taken together, studies on *L. migratoria*, *B. germanica*, and *D. melanogaster* suggest a JH/miRNA/Kr-h1 regulator loop is conserved in insects, but the miRNAs involved appear to be species-specific.

MicroRNAs regulate transitions between alternative developmental pathways

Accumulating evidence suggests that miRNAs also regulate transitions between alternative developmental pathways that generate polyphenic traits. Some well-known examples

of insect polyphenisms that are associated with changes in miRNA profiles include phase transition in locusts [33]; caste determination in honeybees and bumble bees [34–36]; wing dimorphism in aphids, planthoppers, and butterflies [16,37–39]; and facultative diapause in numerous diverse taxa [40].

The transition between long- and short-wing morphs in the brown planthopper (BPH) is currently one of the best-studied examples of miRNA involvement in phenotypic plasticity. Nymphs of BPH can develop into long-wing (LW) or short-wing (SW) adults in response to changes in the environment, especially a change in food quality that affects insulin signaling [38,39]. At the molecular level, the switch between LW and SW is regulated by an autoregulatory feedback loop that includes the insulin/insulin-like growth factor pathway, miR-34-5p, JH, and ecdysone (Figure 2) [38,39]. miR-34-5p targets insulin receptor 1 (InR-1); treating nymphs with a miR-34 inhibitor increases InR-1 translation and promotes development of LW morphs. miR-34 mimics decrease InR-1 translation and results in the production of SW morphs [38]. miR-34-5p expression is regulated by a feedback loop that includes InR-1, JH, and Broad-Complex (Br-C). Br-C is an ecdysone response gene that binds to a promoter region upstream of the miR-34 transcription start site and limits miR-34 transcription [38]. Br-C activity is negatively regulated by JH, and increasing JH titers reduces Br-C activity and allows miR-34 expression to increase. The subsequent decrease in InR-1 allows JH to remain at levels that continue to promote miR-34 expression [38]. Increasing ecdysone titers leads to a subsequent increase in Br-C and downregulation of miR-34 and a shift from the production of LW to SW morphs. A recent profiling study identified additional miRNAs that are correlated with the switch between SW and LW morphs and are likely part of the gene network that regulates phenotypic plasticity in BPH [37].

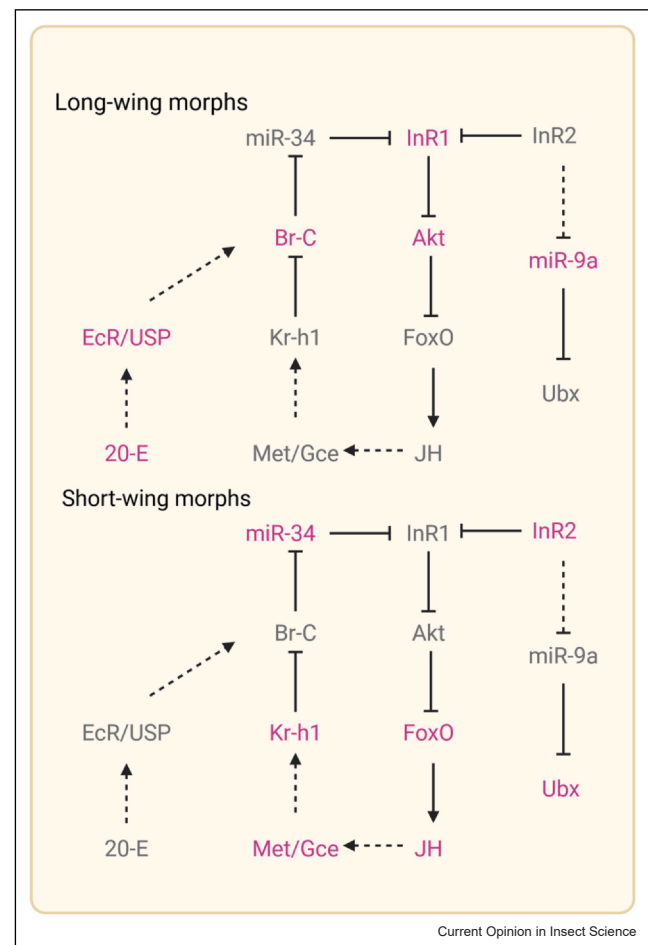
What we are beginning to understand about microRNAs and development

MicroRNAs canalize development

It is becoming apparent that developmental events and transitions are coordinated by GRNs (i.e. collections of genes and molecular mechanisms that regulate expression of those genes). To function appropriately, each GRN needs the ability to quickly recover from any random flux in the system that occurs [41–43]. Accumulating evidence suggests that one function of miRNAs is to stabilize GRNs and prevent unpredicted changes in gene expression from disrupting the developmental program (i.e. canalize development).

A study by Lu et al. [44] provides evidence that miRNAs can canalize development and guard against perturbations from high-temperature stress. A modest 15–20%

Figure 2



MicroRNAs are critical components of regulatory feedback circuits that mediate wing dimorphism in Brown Planthopper. Active factors are labeled in red; inhibited factors are labeled in gray. Figure created using BioRender.com.

Adapted from the paper by Ye et al. [38].

reduction in the pool of mature miRNAs in a Dicer1 knockdown line of *D. melanogaster* did not negatively impact development except for decreasing adult longevity compared to wild-type flies. Increasing the rearing temperature from the normal 25°C to 30°C causes minor development defects in wild-type flies. However, flies from the Dicer1-KO line had substantially higher levels of mortality at all stages of development, especially at the pupal stage. The authors concluded that miRNAs are necessary to buffer the effects of high-temperature stress.

miRNAs may canalize development by acting as switches and effectively turning off proteins that should be restricted to a particular tissue or time in development. One example is the JH/let-7/Kr-h1 regulatory loop described for *L. migratoria* [29]. Another example is an

ecdysone/miR-14/EcR regulatory loop in *D. melanogaster*. EcR is positively regulated by ecdysone and negatively regulated by miR-14 [45]. Transgenic flies that lack miR-14 have elevated EcR titers and a number of developmental defects that can be avoided by reducing EcR production. Transgenic flies that overexpress EcR and normal levels of miR-14 have the same defects as flies lacking miR-14. Together these observations demonstrate that miR-14 functions as a switch to turn off EcR synthesis in *D. melanogaster* [41].

miR-14 also appears to turn ecdysone signaling off following ecdysis in *B. mori* [23]. miR-14-5p expression follows ecdysone titers before and after ecdysis. miR-14-5p abundance is low before ecdysis but is upregulated after ecdysis in conjunction with a steep decline in ecdysone. Inhibiting miR-14-5p after ecdysis is complete keeps ecdysone titers elevated, delays molting, and increases the variance in molting times compared to control larvae. The authors suggest that miR-14-5p canalizes development by synchronizing developmental timing in a population [23].

Processes that microRNAs regulate may be conserved, but the microRNAs involved may not be

Many developmental transitions are regulated by gene networks that include miRNAs, ecdysone signaling, and juvenile hormone signaling pathways. Studies with *D. melanogaster*, *B. mori*, *L. migratoria*, and other insects suggest that such networks are well-conserved in insects and are necessary for development. However, it is becoming clear that the particular miRNAs involved are not necessarily conserved. In *Chilo suppressalis* and *B. mori*, both Lepidopteran species, the ecdysone-biosynthesis protein Neverland is downregulated by miRNAs following ecdysis [10,23]. However, miR-9b targets Neverland in *C. suppressalis*, while miR-14 regulates Neverland in *B. mori*. Inhibiting miR-9b in *C. suppressalis* or miR-14 in *B. mori* disrupts molting. Thus, while miRNAs regulate ecdysone synthesis in both species, the miRNAs involved are species-specific.

There is also evidence that a conserved miRNA may have unique targets and functions in diverse species. In *Tribolium castaneum*, miR-8-3p is necessary for eye development and for the formation of legs and wings. miR-8-3p also regulates key developmental pathways in *D. melanogaster* and *Aedes aegypti* [46,47], and manipulating miR-8-3p in either species causes developmental defects that are similar, but not identical, to those seen in *T. castaneum*. Although these studies demonstrate that miR-8-3p involvement is conserved, gene targets may be species-specific. Additional studies that confirm gene targets of miR-8-3p in more diverse taxa are necessary to demonstrate target and function conservation [21,42,43].

What is needed to demonstrate the function of additional microRNAs in the developmental toolbox?

Our understanding of miRNAs in insect development has increased substantially during the last 30 years. Numerous studies have identified miRNAs that are involved in development and have described changes in miRNA expression that occur as an embryo becomes a mature adult. A large number of studies have also demonstrated that manipulating miRNA abundance through genetic modification or by application of a mimic or inhibitor can cause developmental defects. An increasing number of studies demonstrate that miRNAs are integral parts of gene regulatory networks.

In spite of the new information about miRNAs and insect development, there are still unanswered questions about how miRNAs work with the other tools in the toolbox. For example, how miRNA expression is regulated is not well understood. Identifying promoter sequences and transcription factors that promote or inhibit miRNA expression would provide valuable information about the role of miRNAs in gene networks.

Identifying promoter sequences upstream of miRNA transcription start sites would provide valuable information about miRNA regulation. Another necessary next step is identifying transcription factors that regulate miRNA expression. Finally, discovering where miRNAs are positioned within regulatory networks relative to other genes would add to our understanding of their function. A putative feedback loop that includes miRNAs and Polycomb/Trithorax (PcG/trxG) group proteins has recently been described for *D. melanogaster* [48]. PcG/trxG protein complexes regulate gene expression through their interactions with the chromatin structure. PcG/trxG protein networks regulate ecdysone biosynthesis and are essential for developmental timing [49,50]. Using an *in silico* approach, Solorzano et al. [48] identified Polycomb Response Elements in the *D. melanogaster* genome that may regulate the expression of nearby miRNAs. They also identified PcG/trxG genes that are putative targets of miRNAs. The proposed miRNA-PcG/trxG feedback loops have not been experimentally validated, but they suggest exciting new pathways to explore both in *D. melanogaster* and non-model insects that would add to our understanding of the role of miRNAs in the developmental toolbox.

Data Availability

No data were used for the research described in the article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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